Boston University MF703: Programming for Mathematical Finance

Final Project: Multi-Factor Model

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Introduction

Multi-factor model integrates quantitative data, statistical theories, and financial theories, leveraging a combination of factors to explain and predict asset returns. In this project, we aim to develop a systematic stock selection model based on a multi-factor approach to outperform the market. By identifying and analyzing key drivers of stock performance, this model strives to generate returns through a disciplined and data-driven methodology.

General Process

The project involves the following stages:

1. Data Processing
    - Collect historical stock data and fundamental financial ratios, including P/E, P/B, ROE from Yahoo! Finance API and WRDS.
    - Process and clean the data, ensuring completeness and consistency.
    - Filter out invalid stocks, fill missing values, and organize the data for analysis.
2. Factor Analysis and Quantile Portfolio Construction
    - Explore over 100 potential factors using historical data from 2014 – 2020
    - Standardize, neutralize, and winsorize factors values to ensure comparability.
    - Rank stocks based on factor scores and evaluate quantile portfolio performance metrics, including IC, IR, and p-values.

- Identify the top factors based on their predictive power.

3. Factor Backtesting
   - Use selected factor to construct quantile portfolios with varying rebalancing frequencies (daily, weekly, monthly).
   - Simulate portfolio performance and compare results with market benchmarks, S&P 500.
4. Multi-Factor Stock Portfolio Construction and Backtesting
   - Combine the top three factors (TRIX, Alpha 67, and Alpha 120) to construct a multi-factor portfolio.
   - Select stocks from the top quantile for each factor and assign equal weights.
   - Backtest the portfolio using data from 2021 – 2024 to access performance metrics such as annualized return, volatility, and Sharpe ratio.

Stage 1: Data Processing

In this stage, we seek for data that is potentially meaningful to further research. The primary goal is to collect, clean, and prepare historical stock data and fundamental financial ratio to ensure data consistency, accuracy, and usability for factor analysis and portfolio modeling.

From Yahoo! Finance, we obtained historical data, including open price, high price, low price, closing price, adjusted closing price, and volume. From WRDS, we obtained ratios data, including book-to-market ratio, price-to-earning ratio, price-to-sales ratio, return on asset, return on equity, and price-to-book ratio. Given the extensive data requirement, we had set a target of collecting at least 10 years of historical data to ensure sufficient depth for factor exploration and backtesting. However, due the limitations of yfinance, we only accessed 1253 stocks of total US market stocks. Furthermore, the ratio data frequency from WRDS is monthly.

After obtaining the historical data and ratio data, we set the data as indices of both files, left merge ratio file to historical data file on stock symbol, and forward-fill the ratios. We set multiple error detection mechanisms to ensure the data processing pipeline operated correctly. For instance, we use Try-Except block to handle stocks with varying time ranges, ensuring that discrepancies in data availability did not interrupt the processing workflow.

The final step of data processing involved filtering stock files based on the length of their historical data. Stocks with more than 10 years of history were specifically selected for factor exploration, aligning with the extensive requirements and the need for effective analysis. However, both categories – stocks with histories of more than and less than 10 years – were retained for factor backtesting to ensure comprehensive evaluation across different stock types.

Stage 2: Factor Analysis and Quantile Portfolio Construction

Factor Analysis

In this stage, we aim at constructing factors designed to capture higher returns, primarily by leaveraging momentum-based strategies that align with broader market trends. In other words, we identified and buy stocks that exhibit positive momentum, without short-selling any positions. Since the stock market tends to rise over the long term, and we are testing a broad of stock, we expect these trend-following factors to deliver profits in a long run. After constructing factors, we performed a course of action. First, we use market capitalization to neutralize factors, removing the influence of market cap on factor values. Market capitalization is a well-known systematic driver of stock returns. By neutralizing, we can eliminate market capitalization effects and hedge against market volatility. Second, we standardize to make factor values more comparable. Lastly, we applied winsorization to mitigate the influence of outliers. Extreme values can disproportionately impact statistical measures such as the mean and standard deviation, potentially leading to biased or unstable results. Winsorization ensures that the data remains robust by capping extreme values at predefined thresholds and prevents overfitting issues.

We initially employ a range of well-established momentum indicators, including Relative Strength Index (RSI), Moving Average Convergence Divergence (MACD), 5-Day Moving Average (MA5), and Triple Exponential Average (TRIX), as the foundational momentum factors for testing. In addition to these traditional metrics, we incorporate prevailing factors and methodologies as outlined in publications from leading financial institutions and investment banks, ensuring alignment with industry best practices. Factors can be constructed using absolute values or through derived relationships. For instance, we utilize the relationship between the current closing price and its delayed closing price to develop momentum-related factors. Additionally, factors can be constructed using ratios that capture unique characteristics of the market. As an example, we construct a factor by calculating the ratio of the 10-day mean closing price to the volume-weighted average price (VWAP), providing insights into price trends relative to traded volumes.

Then we calculated the Information Coefficient (IC) table and the Information Ratio (IR) value to evaluate the effectiveness of factors. The IC is typically defined as the Spearman rank correlation or Person correlation between the factor values and the realized future returns over a specific holding period. We focus on identifying effective factors with an IC mean larger than 0.02 or less than -0.02, while also considering the economic rationale and interpretability of the factors to determine whether to apply a momentum strategy or a mean-reversion strategy.

Quantile Portfolio Construction

For each constructed factor, we ranked stocks based on their processed factor scores and divided them into 30 quantiles, from the group with the lowest factor score to the highest. The stocks are equally weighted within each quantile portfolio. This quantile-based sorting allowed us to systematically evaluate the relationship between factor scores and subsequent stock returns.

To analyze performance, we calculated periodic returns and plot histogram at various frequencies, including:

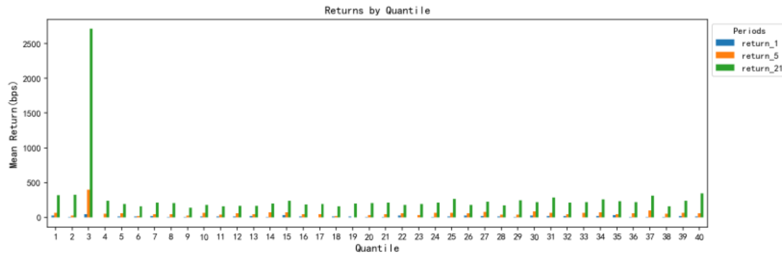
- **Daily Returns:** Capturing short-term factor performance.
- **Weekly Returns (5 trading days):** Providing a medium-term perspective.
- **Monthly Returns (21 trading days):** Assessing longer-term trends influenced by the factors.

By analyzing these periodic returns across different frequencies, we were able to visualize the performance of stocks under each factor. This approach enabled us to identify patterns and trends in stock behavior, highlighting the effectiveness of the factors in predicting returns over varying time horizons. The visualizations provided crucial insights into the consistency and reliability of each factor, further guiding the selection and optimization process.

Based on the criteria, the following three factors were selected for their consistent predictive power and alignment with our objectives:

- *Triple Exponential Average (TRIX)*

  TRIX is a momentum factor that is designed to filter out insignificant price movements and noise, focusing on the underlying trend by applying triple exponential smoothing to the closing prices. The IC table shows that the IC mean of 21 days is close to -0.02. Thus, we take a mean-reversion strategy for this factor, selecting the quantile with the lowest scores.



Return Analysis:

	return_1	return_5	return_21
alpha	0.028626	-0.181904	1.851702
beta	0.911630	1.070637	0.631682

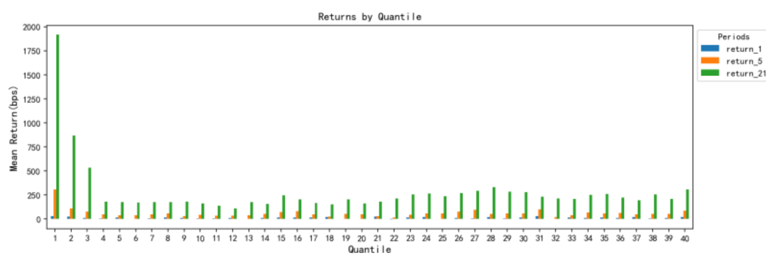
IC Table:

	return_1	return_5	return_21
IC mean	-0.001862	-0.007648	-0.018150
IC std	0.167088	0.169078	0.149883
IR	-0.021123	0.047441	-0.332831
t-stat	-0.430043	-1.745475	-4.672707
p-value	0.667227	0.081109	0.000003
Skew	-0.166049	-0.315424	-0.420728
Kurtosis	0.702885	0.991453	0.300766

- *Alpha 67*

Formula: EWMA(MAX(CLOSE-DELAY(CLOSE,1),0),1/24)/EWMA(ABS(CLOSE -DELAY(CLOSE,1)),1/24)*100

This factor is a momentum factor measuring the proportion of positive price movements relative to total price movement. Empirical tests on a large U.S. equity universe indicate that going long in stocks with lower factor values—indicating less recent upward momentum—consistently delivers excess returns. This suggests the factor is tapping into a mean-reversion or contrarian effect, whereby recently underperforming stocks are more likely to rebound, thus offering a profitable trading opportunity.



Return Analysis:

	return_1	return_5	return_21
alpha	0.034058	-0.231080	1.494734
beta	0.875594	1.085805	0.639223

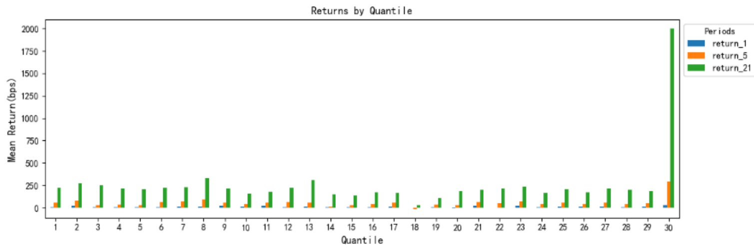
IC Table:

	return_1	return_5	return_21
IC mean	-0.006351	-0.012911	-0.019998
IC std	0.168977	0.175461	0.161328
IR	-0.059564	-0.029035	-0.081195
t-stat	-1.444405	-2.827958	-4.763927
p-value	0.148837	0.004748	0.000002
Skew	-0.171991	-0.471098	-0.460426
Kurtosis	0.529487	0.661456	0.161227

- *Alpha 120*

Formula: (RANK((VWAP - CLOSE)) / RANK((VWAP + CLOSE)))

Alpha 120 is a relative strength factor designed to capture VWAP-based pricing dynamics. It is calculated as the ratio between two percentile-ranked expressions of VWAP and closing price, providing insights into the relative positioning of stock prices. With 30 total quantiles, we believe Alpha 120 effectively differentiates stocks based on their pricing dynamics. Consequently, we apply a momentum strategy, selecting and buying stocks with the highest factor scores.

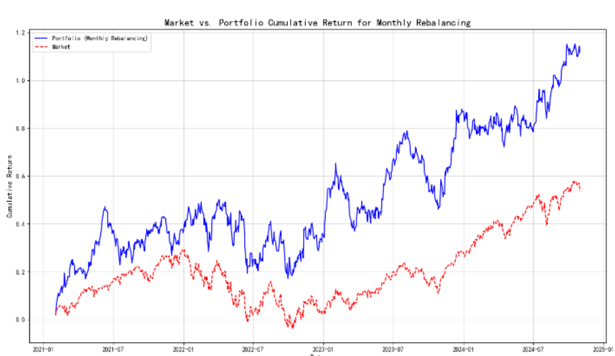


	return_1	return_5	return_21
IC mean	-0.006025	-0.005115	0.006862
IC std	0.200818	0.211156	0.204743
IR	-0.068398	-0.301029	-0.313859
t-stat	-1.152973	-0.931014	1.287980
p-value	0.249108	0.351999	0.197955
Skew	0.201469	0.400213	0.399626
Kurtosis	0.270663	0.433771	0.302992

Stage 3: Factor Backtesting

In this stage, we evaluated and visualized the performance of the constructed factors by simulating their application over the time range from 2021 to 2024. This process aimed to assess the predictive power of the factors and validate their effectiveness in a real-world trading environment. We again employed three rebalancing frequencies consistent with the factor analysis stage: daily, weekly, and monthly. For evaluation, we focused on three key metrics: annualized return, annualized volatility, and Sharpe Ratio.

For example, factor Alpha 120 has following results:



Factor: Alpha 120	Portfolio(Daily)	Market(Daily)	Portfolio(Weekly)	Market(Weekly)	Portfolio(Monthly)	Market(Monthly)
Annualized Return	26.05%	12.71%	23.97%	12.03%	22.66%	12.83%
Annualized Volatility	23.64%	16.60%	23.26%	16.61%	23.02%	16.61%
Sharpe Ratio	1.1	0.77	1.03	0.72	0.98	0.77

The stocks selected based on Factor Alpha 120 demonstrated significantly higher annualized returns compared to the market benchmark across all three rebalancing frequencies (daily, weekly, and monthly). Additionally, the Sharpe Ratio for the Alpha 120 portfolio consistently exceeded the market's Sharpe Ratio, confirming superior risk-adjusted returns.

These results validate that stocks identified by Alpha 120 outperform the market, both in absolute and risk-adjusted terms, supporting the effectiveness of the factor in guiding stock selection and portfolio construction.

Factor Backtesting Results for TRIX and Alpha 67:

Factor: TRIX	Portfolio(Daily)	Market(Daily)	Portfolio(Weekly)	Market(Weekly)	Portfolio(Monthly)	Market(Monthly)
Annualized Return	31.75%	12.71%	29.08%	12.03%	27.22%	12.83%
Annualized Volatility	26.86%	16.60%	26.47%	16.61%	26.62%	16.61%
Sharpe Ratio	1.18	0.77	1.10	0.72	1.02	0.77

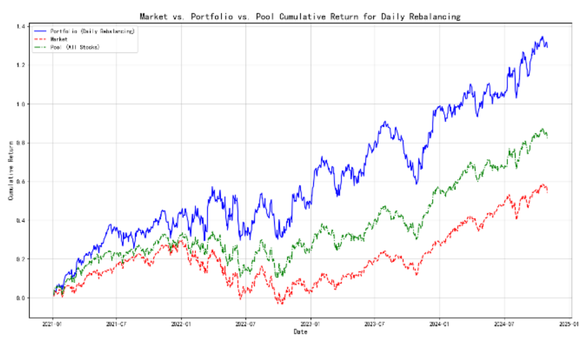
Factor: Alpha 67	Portfolio(Daily)	Market(Daily)	Portfolio(Weekly)	Market(Weekly)	Portfolio(Monthly)	Market(Monthly)
Annualized Return	19.64%	12.71%	19.09%	12.03%	21.70%	12.83%
Annualized Volatility	21.68%	16.60%	20.74%	16.61%	19.89%	16.61%
Sharpe Ratio	0.91	0.77	0.92	0.72	1.09	0.77

Stage 4: Multi-Factor Stock Portfolio Construction and Backtesting

In this stage, we constructed a portfolio using stocks selected by the multi-factor model. The portfolio was built by taking the union of stocks identified by each individual factor, ensuring a comprehensive selection of high-potential stocks. At each rebalancing point, the selected stocks were equally weighted to maintain simplicity and avoid bias toward any single factor.

The portfolio constructed using the multi-factor model outperformed the market benchmark, delivering superior returns as anticipated. Specifically, we compared the multi-factor portfolio return (blue line) against the S&P 500 return (red line). Meanwhile, considering the data limitation and bias of yfinance, we also include the all-stocks-in-pool return (green line). The results consistently demonstrated the portfolio's ability to achieve higher performance. This outcome validates the effectiveness and robustness of the multi-factor approach in driving superior investment returns.

Multi-Factor Stock Portfolio Backtesting Results:



	Annualized Return	**Annualized volatility**	**Sharp ratio**
	Daily		
Portfolio	24.12%	21.99%	1.10
S&P500	12.71%	16.60%	0.77
Pool	17.17%	16.99%	1.01

	Weekly		
Portfolio	22.37%	21.58%	1.04
S&P500	12.03%	16.61%	0.72
Pool	15.96%	16.95%	0.94
	Monthly		
Portfolio	23.31%	21.42%	1.09
S&P500	12.83%	16.61%	0.77
Pool	17.10%	16.97%	1.01

Conclusion

This project successfully implemented a multi-factor model to construct a portfolio that outperformed the market in both returns and risk-adjusted metrics. By leveraging diverse factors such as Alpha 120, TRIX, and Alpha 67, and combining momentum and mean-reversion strategies, we validated the effectiveness of systematic stock selection. The results demonstrate the potential of multi-factor approaches for robust, data-driven portfolio optimization.

References

Darolles, S., & Duvaut, P. (1st Edition). *Multi-factor Models and Signal Processing Techniques: Application to Quantitative Finance*. Wiley, [2011]. (Inspiration for constructing the multi-factor model).

Guotai Junan Securities Research Publications, [Guotai Junan Securities], [2017]. (Sources for factor definitions and methodologies).